



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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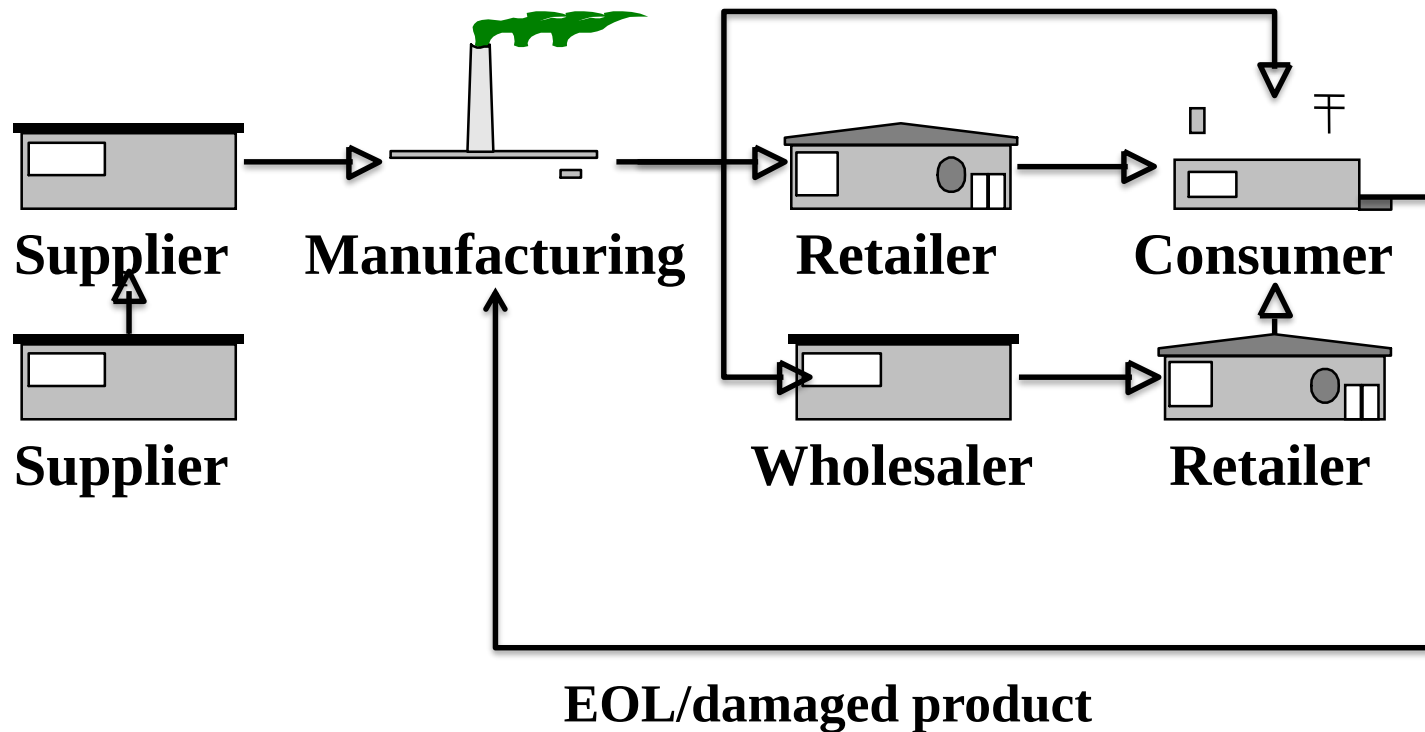


Agenda

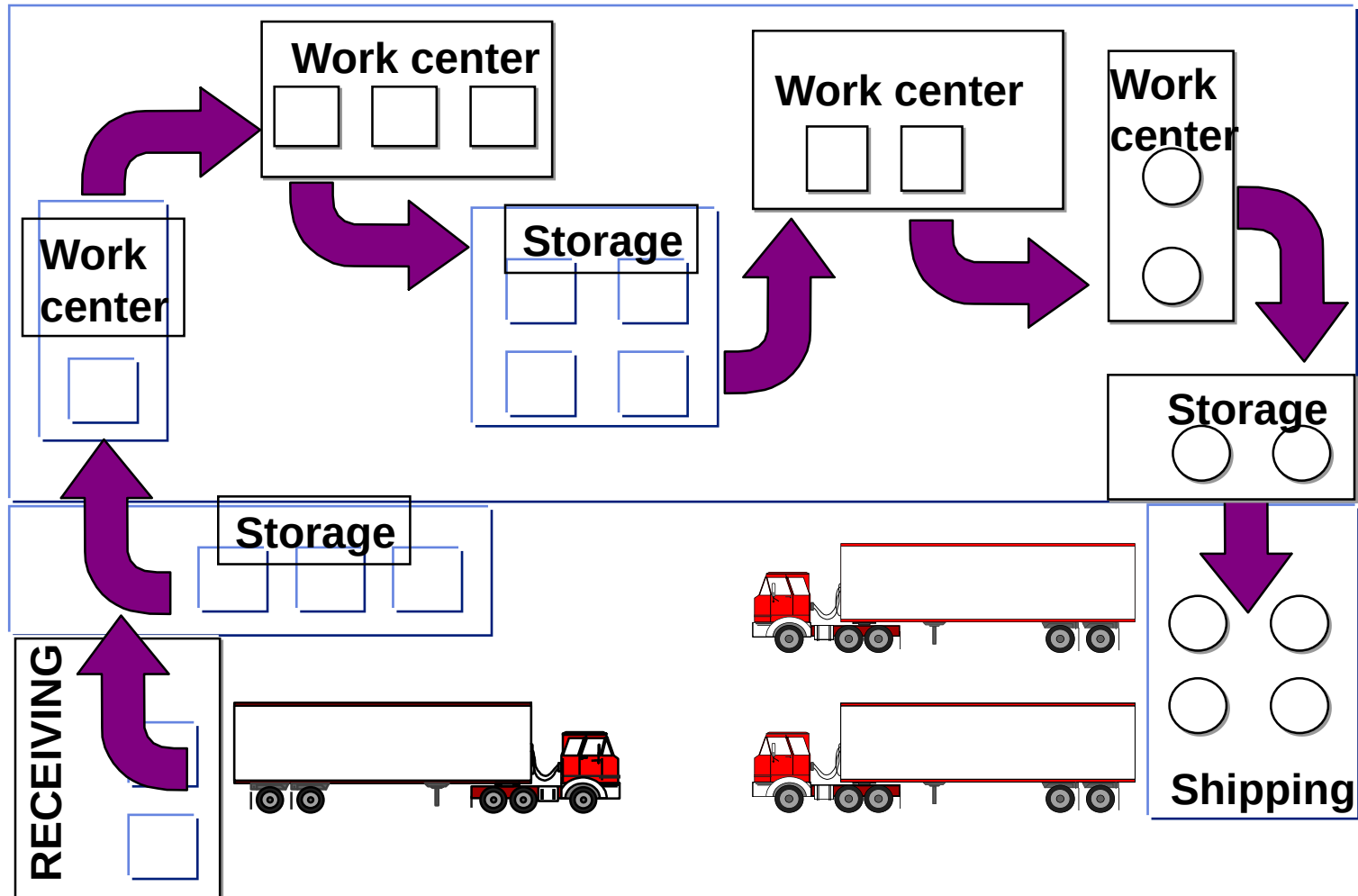
- Chapter 5 Materials Handling, Distribution, and Routing

Material Handling

- To Use Equipment to Move Materials Internally
 - Non-productive (non-value added)



Material Handling



Material handling Equipment

- Conveyors
 - From one fixed point to another
 - Homogeneous material supplied at a constant rate
- Industrial trucks
 - Varying paths
 - Intermittent load
 - Job shops



7-Series Cushion
4-Wheel Electric
3,000-6,500 lbs.



7-Series Cushion
4-Wheel Electric
8,000-12,000 lbs.



Material handling equipment

- Cranes and hoists
 - Overhead lifting devices
- Containers and racks
 - Store and handle bulk material
 - Better use of space



HPLPS-4848



HP2P2-4840



HPLPG-4848-HD
(Not pallet
truck accessible)



HDRUM-4848



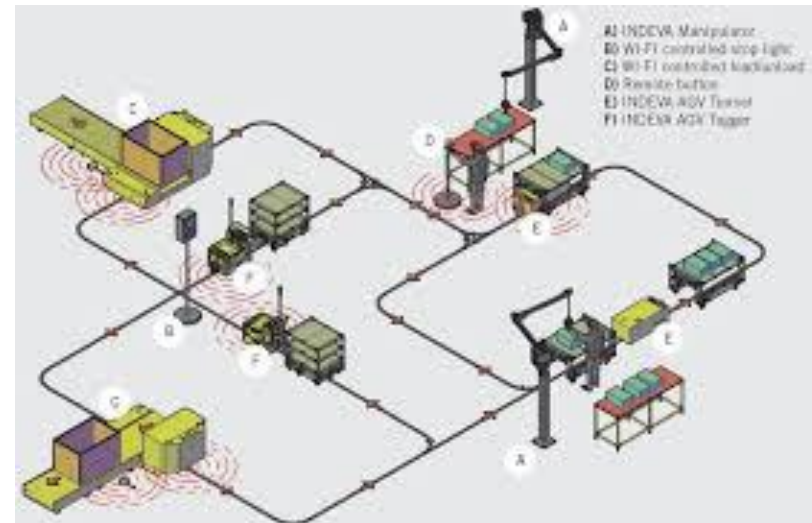
Material handling Equipment

- Elevators and lifts
 - Vertically raise or lower material
 - Fixed in location



Material handling Equipment

- Automatic guided vehicles (AGV)
 - Driverless vehicle
 - Predetermined paths



Material handling Equipment

- Automated storage and retrieval system (AS/RS)
 - Storage rack systems
 - Control system (computer)
 - Crane



Principles of materials handling

- Planning (planned not evolved)
- Systems (material movement cycle)
- Material flow (smooth)
- Simplification (motion economy)
- Gravity (gravity feed bins, roller conveyors)
- Space utilization (high racks with narrow aisle stacking trucks in warehouses)



Principles of materials handling

- Unit size (largest accumulated load, pallets)
- Automation (powered conveyors, automatic pallet stackers)
- Equipment selection (based on all aspects of material and layout)
- Standardization (standard-sized pallets and stacking patterns)
- Adaptability (to perform a variety of tasks, variable-speed conveyors)
- Maintenance (plan for preventive and emergency maintenance)
- Safety (of the operator)

Material routing

- Travelling salesman problem (TSP)
 - Routing a fork truck in a warehouse
- Example 1- the truck must start from the plant (A), visit all warehouses, and return to the plant such that the total travelled distance is minimized.
- One truck can supply all warehouses

From \ To	A	B	C	D	E	F	G
A	-	14	21	20	6	24	9
B	14	-	10	9	9	10	11
C	21	10	-	1	15	9	21
D	20	9	1	-	14	9	20
E	6	9	15	14	-	19	9
F	24	10	9	9	19	-	21
G	9	11	21	20	9	21	-

Traveling Salesperson Problem (TSP)

- A very difficult mathematical problem to solve if the problem size is large
- Optimization or heuristic solution methods
- A simple heuristic: Nearest Neighbor Method
- Other versions of the problem
 - Vehicle routing problem
 - Vehicle has capacity and other limits

Solution to example 1

- Nearest Neighbor Method (feasible route)

Starting from A:

$A \rightarrow E \rightarrow B \rightarrow D \rightarrow C \rightarrow F \rightarrow G \rightarrow A$

Mileage = $6 + 9 + 9 + 1 + 9 + 21 + 9 = 64$

$A \rightarrow E \rightarrow G \rightarrow B \rightarrow D \rightarrow C \rightarrow F \rightarrow A$

Mileage = $6 + 9 + 11 + 9 + 1 + 9 + 24 = 69$

Optimal solution:

$A \rightarrow G \rightarrow B \rightarrow F \rightarrow C \rightarrow D \rightarrow E \rightarrow A$

Mileage = 60

Material routing (Vehicle Routing Problem)

- Multiple travelling salesman problem
 - More than one truck
 - One truck can handle the entire load
- Transportation routing problem
 - One truck cannot handle the entire load
 - Capacity constraints on vehicles
 - Demand satisfaction
 - Heuristic: Clark-Wright procedure

Example 2 – Vehicle routing problem (VRP)

- Supply 6 warehouses (B, C, D, E, F, G) from one distribution center (A).
- Truck capacity: 25,000 units

Traveling Distance matrix

From To	A	B	C	D	E	F	G
A	-	14	21	20	6	24	9
B	14	-	10	9	9	10	11
C	21	10	-	1	15	9	21
D	20	9	1	-	14	9	20
E	6	9	15	14	-	19	9
F	24	10	9	9	19	-	21
G	9	11	21	20	9	21	-

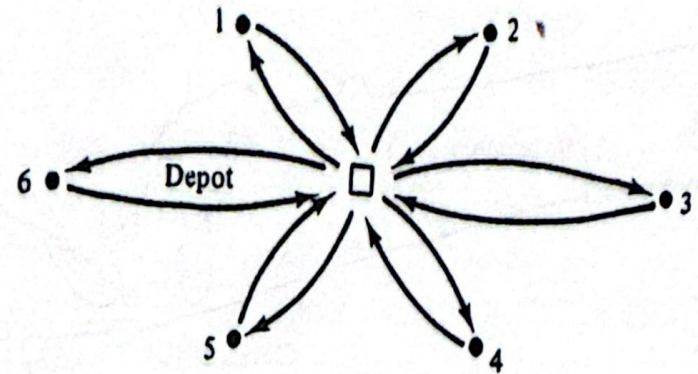
From To	B	C	D	E	F	G
B	-	10	9	9	10	11
C	10	-	1	15	9	21
D	9	1	-	14	9	20
E	9	15	14	-	19	9
F	10	9	9	19	-	21
G	11	21	20	9	21	-

From To	B	C	D	E	F	G
A	14	21	20	6	24	9

	Stop	Demand
G	San Diego	10,000
B	Stillwater	5,000
F	Tampa	6,000
C	College Park	7,000
D	Cleveland	10,000
E	Bozeman	4,000

Clark-Wright procedure

- Initial route generation
- Calculate the savings as the result of pairing nodes
- Rank the savings in a descending order
- Start with the first combination to add to the route and continue on the list until no more stop can be added
 - Truck capacity
 - Route feasibility
 - All destinations are visited



$$S_{ij} = C_{oi} + C_{oj} - C_{ij}$$

C_{oi} = distance from depot to stop i

C_{oj} = distance from depot to stop j

C_{ij} = distance from stop i to stop j

Clark-Wright procedure

Ranked combinations:

<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>
1 CD	40	6 BD	25	11 BE	11
2 CF	36	7 BG	12	12 EF	11
3 DF	35	8 CE	12	13 CG	9
4 BF	28	9 DE	12	14 DG	9
5 BC	25	10 FG	12	15 EG	6

Route

1- Depot-C-D-Depot or vice-versa ($7,000+10,000$)

Depot-F-C-D-Depot or vice-versa ($7,000+10,000+6,000=23,000$)

2- Depot-B-G-Depot or vice-versa ($5,000+10,000$)

Depot-E-B-G-Depot or vice-versa $5,000+10,000+4,000=19,000$)

Total Distance: $24 + 9 + 1 + 20 + 6 + 9 + 11 + 9 = 89 = 8,900$ miles

Solving TSP and VRP problems in various applications

- Easy to formulate the basic versions of TSP and VRP
- Many different and extended versions
- Very difficult to determine the optimal solutions of the general mathematical models
- Simple solution methods may be developed to find good solutions for many practical TSP and VRP problems